

S.E. STRASSMANN

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Interdisciplinary researcher and self-taught coder in human capital economics with training in econometrics, geomatics, computational linguistics, and machine learning. My work addresses measurement gaps affecting marginalized populations — from Indigenous economic valuation to algorithmic fairness in employment systems. I have been directly responsible for the repeal of a federal Canadian immigration policy, contributed to the \$100B Indigenous economy study at Indigenomics Institute, and published on institutional analysis of economic thought. Polyglot fluent in English, French, and Spanish, with classroom training in Urdu, Hindi, and Persian/Farsi.

TECHNICAL SKILLS

Programming: Python (Advanced), R (Advanced), PowerShell/Bash, $\text{\LaTeX}$, SQL (Beginner)

AI / ML: PyTorch (Primary), TensorFlow (Secondary), Transformers, CNNs, RNNs, RL, Deep Learning, Fine-Tuning

Methods: Multimodal AI & Data Visualization (2D/3D/4D, Interactive, Mapping); Geomatics (Vector & Raster Data); Econometrics (OLS, Logit, Probit, Heteroskedasticity Corrections, DiD); Index Creation; History of Economic Thought

Production: Scalable GitHub Projects, Prompt Engineering, Prototypes for ML Applications

Languages: English, Spanish, French (fluent); Urdu, Persian/Farsi (classroom)

EDUCATION

Université de Montréal

Fall 2026

M.A., Political Science — Research stream (with thesis)

- Admitted March 18, 2026 (offer accepted). Tutor: Prof. Jean-François Godbout.
- Research focus: political economy of methodological adoption in social science disciplines.

McGill University

2023

B.A., Economics — Data Science Concentration

- Completed Masters/PhD-level coursework under Prof. Elissa Berwick in quantitative social science.
- Self-taught coder using Compute Canada supercomputing resources for independent research.
- Strategically customized degree around mixed honours coursework across economics, political science, geomatics, and linguistics. Minor: Urdu Language.

CIFAR & First Nations University of Canada

2024

Certificate — Reconciliation Education & Indigenous Perspectives in AI

Center for AI and Digital Policy (CAIDP)

2025

Certificate — AI Policy (with Distinction)

Studio 1

2025

Certificate — QcSE: Scientific Entrepreneurship

RELEVANT EXPERIENCE

AI4Good Lab Fellow × Mila ELab

May 2025 – Nov 2025

Mila — Quebec Artificial Intelligence Institute

- Founder of Mindful Human Capital; addressing methodology gaps in economics training.
- Led an interdisciplinary team developing a foundational model of the brain — multimodal AI for brain encoding prediction using Transformers, CNNs, and RNNs on audio, text, and video.

- Formal accelerator-style education from senior engineers at Mila and Google DeepMind: mathematical foundations, supervised/unsupervised learning, CNNs, RNNs, transformers, and reinforcement learning.

Data Specialist & Executive Support

Sep 2024 – May 2025

Indigenomics Institute & Global Centre for Indigenomics

- Operating at the heart of the economic reconciliation movement, introduced innovative frameworks quantifying Indigenous business contributions by integrating traditional knowledge with modern economic tools.
- Core technical contributor to the \$100B Indigenous Economy AI Study — reporting to Carol Anne Hilton and AI Advisory Chair K Krasnow Waterman.
- Introduced data sovereignty frameworks; recommendations allowed the study to proceed without settler/colonial data and to include non-English-language sources.
- Produced an estimate **3× higher** than Statistics Canada's prior figure for Indigenous contribution to GDP.

Distinguished Research Member

Jan 2025 – Mar 2025

Center for AI and Digital Policy (CAIDP)

- Country-section audit and analysis published in the internationally recognized *AI & Democratic Values Index 2025* (AIDV 2025).
- Mapped data-governance ecosystems, legislation, and stakeholder dynamics in AI policy.
- Translated technical AI findings into strategically relevant insights for international AI policy briefs.

Researcher

Apr 2019 – Aug 2023

Cross-Institutional Research, McGill University — Multi-location

- Sourced funding and led studies leveraging supercomputing resources across language economics, housing, and employment.
- Rotated departments, methods, and advisors to build multimodal expertise.
- Applied quantitative and qualitative methodologies to evaluate public policy with measurable impact.

KEY PROJECTS

Canadian Human Rights | *Federal immigration policy, inclusive pedagogy, and journalism*

2025

- Directly contributed to the repeal of a Canadian federal immigration policy without formal authority or Canadian voting rights. *Medical Inadmissibility* no longer violates the *Canadian Charter of Rights and Freedoms*.
- Effected change on real immigration cases within a month of intervention.
- Communicated research findings on reporting bias to a lead editor at a major US media outlet; directly responsible for content and narrative shifts in featured stories (prime-time radio) within months.

\$100B Indigenous Economy AI Study | *Quantitative Indigenous research methods*

2025

- Recommendations directly enabled the study to be conducted without settler/colonial data.
- Resulted in a 3× higher estimated total Indigenous contribution to GDP, quoted by Statistics Canada.
- Geomatics background allowed activity to be mapped to the land; recommendations ensured capacity for economic reporting from corpora written in native languages (e.g. Inuktitut).

Project Telepath | *Multimodal AI, Supercomputing, PyTorch, and Transformers*

2025

- Developed encoder-decoder architecture predicting fMRI signals from synchronized audio, text, and video data.
- Implemented feature extraction, classification, and prediction pipeline: OLS, BERT, ViT, and ImageBind.

PUBLICATIONS & WORKING PAPERS

Strassmann, S. (2026a). *Human capital of linguistic diversity*. Working paper — access to requisite testable data available on request.

- Strassmann, S. (2026b). *It's of little use: Crying over spilled utility functions*. Working paper — viewable upon publication.
- Strassmann, S. (2023a). *Ability and neurodiversity*. OSF Preprint. <https://doi.org/10.17605/OSF.IO/T4E37>
- Strassmann, S. (2023b). A replication of Chiswick & Miller (2007): Language skills, linguistic concentrations, and the business cycle. *OSF Preprint*. https://osf.io/u5cxn/?view_only=cb3c5d6041fd4d31a5a8d4a518236d34
- Strassmann, S. (2023c). *Waiting on Superwoman: Researcher-centric methodology*. OSF Preprint. https://osf.io/astzq/?view_only=71d5e67d8a3a4d0bbb4bb7c3d027302c
- Strassmann, S. (2023d). *Rethinking public transportation for the neurospicy*. OSF Preprint. https://osf.io/5ghb6/?view_only=500c49a77d664b64a08f709382c878f5
- Strassmann, S. (2023e). *Researcher-oriented methodology for GIS*. Working paper — viewable upon publication.
- Strassmann, S. (2022a). Is it time for Legault to LeStop? The language economics of Bill 96. *McGill International Review*. <https://www.mironline.ca/is-it-time-for-legault-to-lestop/>
- Strassmann, S. (2022b). John Stuart Mill on the merits of colonization. *McGill Journal of Economics*. <https://doi.org/10.17605/OSF.IO/J4BV9>
- Strassmann, S., Benavides-Alfaro, P., Sarr, K., Pollack-Toro, S., & Bergeron, S. (2020). Integrating Indigenous perspectives in environmental policy: Bridging gaps for effective reconciliation. *Working paper*.

CONFERENCES & AFFILIATIONS

- *Leadership and Neurodiversity: Embracing Strengths in the Workplace*. 4th Space, Concordia University, Montreal, Dec 2024.
- *Towards Substantive Equality in AI*. GPAI, CEIMIA & Mila — Quebec AI Institute, Montreal, Nov 2024.
- *Indigenomics NOW*. Global Centre of Indigenomics, 24h international virtual conference, Nov 2024.
- *CIREQ Interdisciplinary PhD Student Conference on Big Data and AI*. CIREQ, Montreal, Jun 2023.
- *Choosing the Right Target: Real Options for the Bank of Canada's Mandate Renewal*. Max Bell School of Public Policy, McGill University, Sep 2020.
- *Better Policy for a Better World: Why Good Ideas Aren't Enough*. Max Bell School of Public Policy, McGill University, Nov 2019.

ACHIEVEMENTS & RECOGNITION

Awards: Startup Community Award Nominee; Real Ventures Fellow (leading VC in Canada); McGill Arts Internship Award (2×); Tomlinson Engagement Award for Change; Youngest university recipient of BLUE Fellowship (McGill's Building 21); 94th percentile, US National Spanish Exam; 34/36 ACT Score (English).

Volunteering: Perfect user rating as trained crisis support & active listener, 7 Cups (multilingual, pandemic). Research mentor and coach, Building 21 at McGill University. Volunteer buddy, McGill University International Student Services (ISS). Economics content moderator, Facebook economics community (anti-disinformation).